

Funding Support to I-STEM Members

Objectives & Benefits:

Researchers may convert their ideas into innovations; develop and fabricate R&D Equipment/ Products /Technical Consumables Indigenously to make India "Aatma Nirbhar (Self Reliant)"

Objectives & Benefits:

1. This program will help in maintaining R&D equipment beyond the Project Period, if they are actively shared with external users through the I-STEM portal
2. Such maintenance services will extend the life of expensive imported & Indigenous equipment, while also reducing tool down time.
3. Easy handling of maintenance through Suppliers and Service Providers listed on I-STEM

Partial Support to Academic Users/ Start-up's and Custodians

Skill Development Training to Prepare Tool Engineers and Consultants for Various R&D Labs

Objectives & Benefits:

1. To fulfill the needs for the Skilled manpower in various R&D Labs for Operation and Maintenance of Equipment
2. Trained tool engineers may start their own firms and support needy scientific Labs (as and when needed, in slot-wise manner, similar to how doctors practice in hospitals)
3. Enhancing Employment Opportunities for the Scientific community

Support for Comprehensive Annual Maintenance of R&D Equipment

Partial Financial Support to Researchers from Academic Institutions/ Start-ups

Objectives & Benefits:

1. Researchers may carry out their R&D/PhD thesis work in labs across the country on a "Pay & Use" basis, and get reimbursed for Travel, Consumables and Lab use charges.
2. Researchers may start collaborative work with various Institutions and Establish a hub of Scientists/Experts to resolve regional issues adopting a scientific approach.



For more details:

<https://www.istem.gov.in/rd-infrastructure-map/funding-opportunities-researchers>

Software Services through I-STEM at No Cost



Conduct meetings/webinars through I-STEM at No Cost



COMSOL Multiphysics 5.6 through I-STEM at No Usage fee

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LAVO e-bike (Credit: <https://www.electronicsforu.com>)

for e-bikes. With the help of solar energy, LAVO's hybrid hydrogen battery operates to generate energy from the extraction of hydrogen from water. The hydrogen is then absorbed into a patented metal hydride, which converts it into battery power.

Battery-free wearable device measures nicotine exposure

Researchers from prominent Australian and the US research institutes have developed a battery-free, wearable device that helps e-cigarettes smokers to analyse in real time their



This flexible nicotine sensor attaches to skin, continuously measuring the wearer's exposure (Credit: www.acs.org)

nicotine consumption and measure the extent of acquiring cardiovascular and respiratory disorders. By choosing vanadium dioxide (VO_2) on a polyimide substrate as the basis for the sensor, it was shown that nicotine can bond covalently to a thin film of VO_2 , thereby altering the film's conductivity depending on nicotine concentration, detecting the change in conductivity, and wirelessly transmitting it to a smartphone. The device can also help non-smokers to gauge the level of exposure to second-hand smoke.